Finite Element Analysis of Magnetic Circuits

1. Introduction to Finite Element Analysis (FEA)

What is FEA?

Finite Element Analysis is a numerical method used to find approximate solutions to boundary value problems for differential equations. In electromagnetics, we use it to solve Maxwell's equations across complex geometries where analytical solutions (hand calculations) are impossible.

How it Works:

1. Discretization (Meshing): The software divides the geometry into thousands of small, simple shapes called "elements" (e.g., triangles or quadrilaterals in 2D). Together, these form the mesh.
2. Local Equations: Within each element, the software assumes a simple variation of the field (usually linear or quadratic).
3. Assembly: The individual equations for every element are combined into a massive global matrix equation.
4. Solution: The computer solves this system of linear or nonlinear equations to determine the magnetic field at every node.

2. Step-by-Step Magnetic Circuit Construction

1) Model Wizard Setup

- Open COMSOL and select Model Wizard > 2D.
- Select AC/DC > Electromagnetic Fields > Magnetic Fields (mf). Click Add.
- Select Study > Stationary. Click Done.

2) Parameters

I. Under *Global Definitions*, define I_test (Current) and N_turns (Last two digits of your Student ID).
   Assign:
   - Name: I_test, Expression: 10[A], Description: Excitation Current
   - Name: N_turns, Expression: Last two digits of your Student ID, Description: Winding turn

3) Setting up the Geometry

I. Left-click on the Geometry node
   i. Under Settings, Change the *Length Unit* to 'cm'.

II. Setting up the surrounding air environment
   i. Locate Geometry1: Right-click on the Geometry 1 node.
   ii. Select Rectangle.
   iii. Lable it Air
   iv. Define Dimensions: In the Settings window for the rectangle:

a. Set the Width to `200 cm`.
b. Set the Height to `200 cm`.

v. Positioning: Under the Position section:
   a. Set Base to Center.
   b. Ensure both x and y coordinates are 0. This centers the "environment" box around the origin.

vi. Build: Click the Build Selected button at the top of the Settings window. You will see a large square appear in the Graphics window.

The magnetic core is created by drawing two squares and subtracting the smaller one from the larger one.

III. Creating the Magnetic Core (Outer Square)

i. Add Outer Square: Right-click Geometry 1 again and select Rectangle.
ii. Lable it Outer Square.
iii. Define Dimensions: In the Settings window:
   a. Set the Width to 50 cm.
   b. Set the Height to 50 cm.
iv. Positioning: Under the Position section:
   a. Set Base to Center.
   b. Ensure x and y are 0.
v. Build: Click Build Selected. You will now see a smaller square inside the first one.

IV. Creating the Core Hollow (Inner Square)

i. Add Inner Square: Right-click Geometry 1 and select Rectangle.
ii. Lable it Inner Square.
iii. Define Dimensions: In the Settings window:
   a. Set the Width to 30 cm.
   b. Set the Height to 30 cm.
iv. Positioning: Under the Position section:
   a. Set Base to Center.
   b. Ensure x and y are 0.
v. Build: Click Build Selected.

V. Subtracting the Inner Square (Creating the "Ring")

This step uses a "Boolean Operation" to cut the hole out of the middle, leaving a square core with 10 cm thick limbs.

i. Select Difference Tool: Right-click Geometry 1, go to Booleans and Partitions, and select Difference.
ii. Objects to Add: In the Objects to add section of the Settings window, click the Outer Core (the 50 cm square) in the Graphics window. Its name (r2) should appear in the list.
iii. Objects to Subtract: Click the Objects to subtract selection box so it turns active (gray/green), then click the Inner Core (the 30 cm square) in the Graphics window. Its name (r3) should appear in the list.
iv. Final Build: Click Build Selected.

VI. Primary Coil_1:

i. Add Primary Left Winding: Right-click Geometry 1 and select Rectangle.
ii. Lable it Primary Left.
iii. Define Dimensions: In the Settings window:
   a. Set the Width to 4 cm.
   b. Set the Height to 20 cm.
iv. Positioning: Under the Position section:
   a. Ensure the Base is set to Corner.
   b. Set x= -30 and y= -10.
v. Build: Click Build Selected.

VII. Primary Coil_2:

i. Add Primary Right Winding: Right-click Geometry 1 and select Rectangle.
ii. Lable it Primary Right.
iii. Define Dimensions: In the Settings window:
   a. Set the Width to 4 cm.
   b. Set the Height to 20 cm.
iv. Positioning: Under the Position section:
   a. Ensure the Base is set to Corner.
   b. Set x= -14 and y= -10.
v. Build: Click Build Selected.

VIII. Final Build: Click Build All Objects. You should now see a large 200 cm box (the environment) containing a hollowed-out square core (the magnetic circuit). Plus, the winding boxes. The simulation is now ready to add the coils and assign materials!

4) Setting Up the Materials

I. Assign the Environment (Air)

- Locate Materials: Right-click the Materials node in the Model Builder and select Add Material from Library.
- Search for Air: Under the "Built-In" folder, select Air and click Add to Component. In this step, the software assumes all the domains to be air.
- Selection: In the Geometric Entity Selection window, select the large outer domain (usually the box or sphere containing your model).

II. Assign the Core (Soft Iron)

- Add Iron: In the same "Add Material" window, Under "AC/DC" library, select ‘Soft Iron(without losses)’ and click ‘+Add to Component’.

- Selection: Click on the core of the magnetic circuit. Domain 3 is added in 'Geometric Entity Selection' of Soft Iron settings.
- Go back to the settings of 'Air', you will see Domain 3 is overridden.
- Click on the Soft Iron material node you just added. Click on the B-H Curve. Click on the Interpolation. You will see a table showing the BH characteristics of the used iron. Click on the 'Plot'. The B-H curve is shown.

5) Setting Up the Physics

I. Inspecting the Magnetic Fields Interface

i. Interface Selection: In the Model Builder, click on the Magnetic Fields (mf) node.
ii. Domain Verification: Review the Domain Selection window. By default, all domains should be included, indicating that the magnetic field equations will be solved throughout the entire geometry (both the core and the surrounding air).
iii. Maxwell's Equations: Expand the Equation section. Here, you can observe the specific vector form of Maxwell's equations that the software employs for this study. This confirms that the simulation is calculating the magnetic vector potential (A) based on your material properties and current inputs.
iv. Thickness Settings (2D): Locate the Thickness field under the 2D Approximation section.
    a. The value is set to 1 m by default. Change this to 0.10 m.
    b. This implies the software treats the 2D cross-section as being extruded for 0.1 meter in the direction perpendicular to the screen (z-direction). It becomes the depth of the magnetic circuit.
    c. For this simulation, 1 m is appropriate as it simplifies the calculation of total magnetic flux and inductance.
v. Before proceeding with custom configurations, examine the default nodes under the Magnetic Fields (mf) interface to understand the underlying assumptions of the model:
vi. Free Space (Ampère's Law): Click this node to see the domains representing the air and the core. Notice that by default, the software applies Maxwell's equations for magnetic fields. Check the Equations section to see how it calculates magnetic flux density (B) and magnetic field intensity (H).
vii. Click the Magnetic Insulation 1 node to review the boundary selections. By default, the software applies this condition to all exterior boundaries of your geometry, enforcing that the magnetic field remains contained within the simulation space. This setting highlights the critical importance of your environment domain dimensions. Because this condition physically forces the magnetic flux to zero at the boundary, the "air box" must be sized significantly larger than the magnetic circuit itself. If the boundaries are too close to the core, the simulation may produce inaccurate results by artificially "squeezing" the magnetic field. A sufficiently large environment ensures that any leakage flux naturally decays to near-zero before reaching the insulated boundary, resulting in a more realistic simulation.

viii. Initial Values 1: Review this node to see the starting conditions. Usually, all field values are set to zero (0) to provide a clean slate for the solver to begin its iterations.

Note: At this stage, except for setting the core depth, no other changes are required. Use this time to familiarize yourself with how the software maps mathematical equations to your 2D geometry.

II. Assigning Ampere's Law Physics

i. Ampere's law in the air

i. Right Click on Magnetic Fields, on More Domains, Select 'Ampere's Law in Fluids'

ii. Under Domain Selection, Select All Domains.

ii. Ampere's law in the core

a. Right Click on Magnetic Fields, on More Domains, Select '**Ampere's Law in Solids**'
b. Under Domain Selection, Select Domain 3 (Which is the core).
c. Under Constitutive Relation B-H, Magnetization model, Select B-H Curve.
d. Go back to 'Ampere's law in Fluids' and make sure that the Domain 3 is overridden.

III. Assigning Coil Setting

i. Right Click on Magnetic Fields, Select 'Domain Coil'

ii. Under Domain Selection, Select 2 and 5 which are the domains assumed to be coil location.

iii. Under Conductor Model, Select: Homogenized multiturn, check the group coil box.

iv. Make sure that the Coil Excitation is 'Current'.

v. Assign Coil Current to be 'I_test'. If you have introduced this parameter in the parameters section, the color of 'I_test' becomes black which means the software identifies the parameter.

vi. Assign Number of turns to be 'N_turn'.

vii. You should make sure that the current in left rectangular is reversed to the right one. To do so, Right click on the 'Domain Coil 1', Select 'Reversed Current Direction'. Under Domain Selection, Select 5.

6) Setting Up the Mesh

i. Select Mesh Node: Click on Mesh 1 in the Model Builder.

ii. Adjust Element Size: Under the Mesh Settings, locate the Element Size dropdown and select Extremely Fine.

iii. In the 'messages' window, you will see this message: Complete mesh consists of 26654 domain elements and 637 boundary elements.

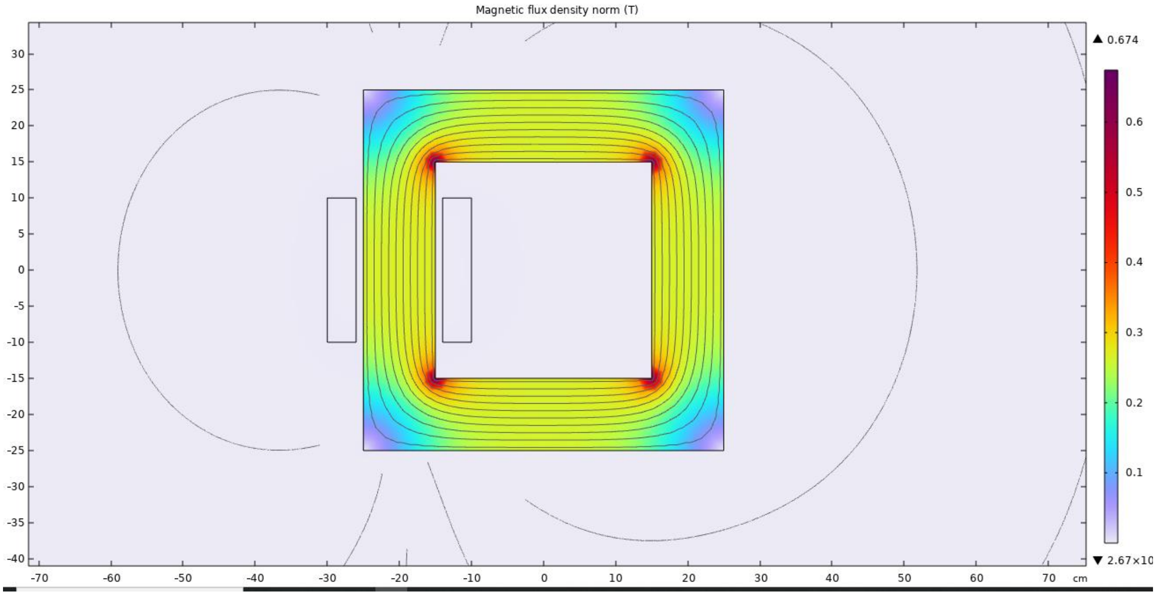
iv. Please note that this setting determines the discretization of your geometry. A finer mesh reduces the numerical error and enhances the accuracy of the results. While "Extremely Fine" provides high precision, it significantly increases the number of degrees of freedom. For very complex or large-scale 3D models, using such a dense mesh can lead to excessive computation times or "Out of Memory" errors. In those instances, a coarser mesh or a "User-Controlled Mesh" (with refinement only in areas of high flux density) is preferred to balance accuracy with available hardware resources.

7) Setting up the 'Study'

I. Execute the Solver: In the Model Builder, select the Study 1 node and click the Compute button in the top toolbar.

8) Results

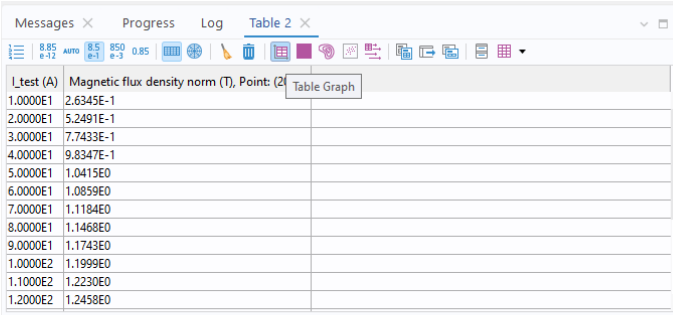
I. View Results: Once the calculation is complete, navigate to the Results section and select the Magnetic Flux Density (mf) plot.
II. Analyze the Field: The graphics window will display the distribution of the magnetic flux density across the iron core. You should observe higher flux concentrations within the core compared to the surrounding air.
III. Important Note on Values: Please be aware that your specific numerical results (measured in Teslas) may differ from the example figure or your classmates' work. This variation is expected because your Number of Turns (N) is based on your unique student ID digits, which directly impacts the magnetomotive force (MMF).



IV. In the **Results** section, right-click on 'Datasets' and select **'Cut Point 2D'**. In the settings window, enter **(20, 0)** as the (x, y) point data. Click **Plot** and verify that the point is positioned at the center of the core's right limb.
V. In the Results section, right click on 'Derived Values'. Select 'Point Evaluation'. Under Settings, in the 'Data' section, assign Dataset as 'Cut Point 2D 1'. In the 'Expressions' section, click on +, click on 'magnetic fiels', click on 'magnetic', click on 'mf.normB'. Click on 'Evaluate'. The calculated value is shown in Table 1.

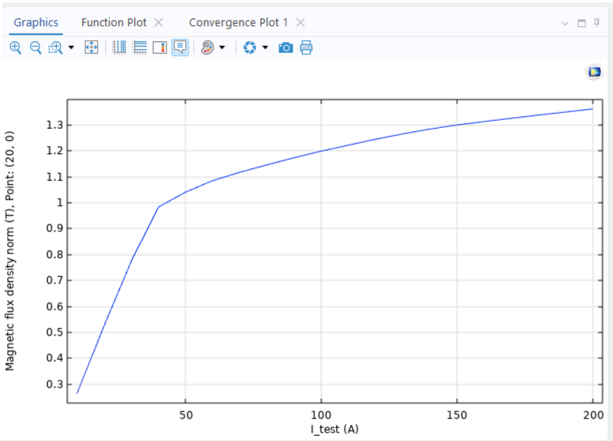
9) Right Click on Study 1. Select 'Parametric Sweep'. Under 'Study Settings', click on +, I_test appears as a parameter. In the 'Parameter value list', type 'range(10,10,200)'. Click on 'Compute'.

10) In the 'Results' section, expand 'Derived Values', click on 'Point Evaluation 1'. Click on 'Evaluate'. Table 1 shows the calculated values. Select 'Table Graph' to plot the magnetic field density based on excitation current.



I_test (A)	Magnetic flux density norm (T), Point: (2(
1.0000E1	2.6345E-1
2.0000E1	5.2491E-1
3.0000E1	7.7433E-1
4.0000E1	9.8347E-1
5.0000E1	1.0415E0
6.0000E1	1.0859E0
7.0000E1	1.1184E0
8.0000E1	1.1468E0
9.0000E1	1.1743E0
1.0000E2	1.1999E0
1.1000E2	1.2230E0
1.2000E2	1.2458E0

The plot:



3. Magnetic Circuit with Air Gap.

I. **File Preparation:** Create two separate simulation files by saving the existing model first as **'Non-linear Magnetic Circuit'** and subsequently as **'Magnetic_Air Circuit'**.

II. II. **Model Modification:** Modify the second model by inserting an air gap measuring 5 mm in length into the middle of the core's right limb.

III. **Analysis Update:** Re-run the plot analysis on the modified model. The previously defined point at coordinates (20, 0) must be at the midpoint of the new air gap.

IV. **Data Comparison & Reporting:** Extract the magnetic flux density (B) versus excitation current (I) data from the plots of both models. Generate a comparative plot using MATLAB or Excel and provide a discussion on the observed effects of the air gap.

4. Design a simulation scenario that demonstrates magnetic flux leakage. Run the simulation and report your findings with supporting plots or calculated values.

5. Design and simulate a scenario that demonstrates the magnetic fringing effect around an air gap. Report your results with supporting field plots.